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Report on AI use for Lab Work 1

Information and Coding (2026/27)

Work done by:
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AI Usage by Team Member

José Fernandes (114472)

Used AI to help with some Go syntax, clarify some concepts mainly associated with the lossless codecs, and document formatting in the report.

- **Part II, Exercise 5 (Encoder Logic):** AI was queried to help refactor bit-level operations into byte-level operations. Because Go does not natively support addressing individual bits. It was correct and was checked with practical testing. Learned how effectively use []byte arrays for faster processing.
- **Conceptual Understanding:** AI was used as an interactive search tool to better understand the structural concepts behind Huffman coding before implementing the algorithm. This was done side by side with regular searching methods like the Go documentation and forums.
- **Report Formatting:** AI assisted in generating LaTeX "objects", specifically formatting the structural code for tables and writing a good base for standard glossary definitions that were later altered to keep simplicity and consistency along the whole report. The template for the AI report was also partially done with AI. No critical knowledge or conclusions were generated with AI.

Diogo Carvalho (113221)

Used AI to understand the Huffman tree and priority-queue code and to speed up the decoder implementation, avoiding repetitive work mirrored from the encoder, as well as to help automate benchmarking.

- **Part II, Exercise 5 (Encoder):** AI was used to understand the existing code. While most of the planning was done by me, AI helped validate my plan and helped correct some of the code blocks after I wrote them, to avoid wasted time debugging small bugs in code rather than in logic, though it also wrote some small chunks of code I got stuck in. It also helped clear up a few doubts I had with canonical Huffman coding, especially given some particular Go behaviours.
- **Part II, Exercise 5 (Decoder):** AI was used here to draft a rough plan on how the decoder should be developed, given the already completed encoder. It was subsequently used to verify the code, as well as for helping rewrite code that was somewhat close to its encoder counterpart.
- **Benchmarking:** AI was used to generate a script based on a few requirements and a few already run commands, to speed up the repetition (per file), measuring and logging of the entire benchmarking process. It was also useful as I'm not entirely familiar with how to measure memory for a single process on macOS.

Rúben Franco (114827)

Did not use AI tools at any point. All of his coding, research, and documentation tasks were completed using traditional methods, official language documentation, and manual implementation.

Conclusion

The integration of AI in this project was kept mostly to a supportive role, exceptionally used more actively to dramatically speed up writing code in later parts of development, especially as code was being worked on by different people. Design decisions remained ours and were discussed in group when appropriate. All suggestions were checked against the course and project notes, and all code was manually tested and shared with the other members for further analysis. The AI tools used were usually correct, and were useful in anticipating problems and suggesting better alternatives, but they were most valuable for finding holes in our logic when trying to understand the topics at hand.